

Multi-hop Relay Directional Communication Based on Attractants in Molecular Communication

Hao Du
Yangtze Delta Region Institute
(Quzhou), University of Electronic
Science and Technology of China
Quzhou, Zhejiang, China
dh_duhao@163.com

Qiang Liu*
University of Electronic Science and
Technology of China
Chengdu, Sichuan, China
liuqiang@uestc.edu.cn

Kun Yang
University of Essex
Colchester, Essex, UK
kunyang@essex.ac.uk

ABSTRACT

In this paper, a new multi-hop relay algorithm based on three attractants is proposed for the traditional multi-hop relay model. When the nano-machine encounters different attractants, the algorithm allows the nano-machine to choose the direction with a higher concentration of the attractant by using the relative priority between the attractants. The efficiency of the algorithm is greatly improved.

CCS CONCEPTS

• **General and reference** → **General conference proceedings**; *General literature*; Experimentation.

KEYWORDS

Molecular communication, chemotaxis, multi-hop relay, relative priority, target detection

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1 INTRODUCTION

In order to solve the problem of the low efficiency of nano-machines in finding targets. So far, we have adopted the method of using fixed three different attractants, and the relative priority between different attractants to help the nano-machine to the target. While reducing the types of attractants, it maintains good performance.

Some related works are introduced as follows. In [2], due to the random diffusion of molecules, chemotactic substances can not quickly reach the remote nano-machine, which affects the effectiveness of detection. [1] proposed a multi relay algorithm based on [2]. Different relays use different attractants to study the impact of adding such relay nodes on the performance of directional communication control algorithm.

*Corresponding author

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2 SYSTEM MODEL

In this section, we assume the following for targets and nano-machines (including relay nodes).

- There is a fixed target in the environment and it is located in the center of the environment.
- The target releases attractant molecules.
- Nano-machines can detect the presence of the target when they are in close proximity to the target.
- Nano-machines can detect different attractive molecules, and nano-machines can move, stop.

In this model, the relay can release three different kinds of attractant molecules B_1, B_2, B_3 , but only one attractant molecule can be released at the same time, and the target releases the attractant molecule A. The method of priority allows the nano-machine to select one of the attractants and to move chemotactically in the direction of its higher concentration.

Algorithm 1 shows the movement strategy of the nano-machine, and Algorithm 2 shows the release attractant strategy and the movement strategy of the relay node.

2.1 Concentration Of Type A, B_i Molecules

The diffusion of type A and B_i molecules on Fick's diffusion law. The concentration formula of A and B_i is

$$\frac{\partial C(X, t)}{\partial t} = D \nabla^2 C(X, t) + r(X, t) \quad (1)$$

According to the above formula, the concentration of A at position X and time t can be calculated as:

$$C(X, t) = \frac{M}{4\pi Dt} \exp\left(-\frac{|X_T - X|}{4Dt}\right) \quad (2)$$

Where D is the diffusion coefficient of the molecules, and $r(X, t)$ is the rate at which the target produces the molecules. $mK \leq t \leq (m+1)K$, $m = 0, 1, 2, \dots$, where m represents the m -th time, K is the time interval for the target and relay to release A, B_i , X_T represents the coordinates of the target and relay, M represents the parameters related to the amount of A, B_i released by the target and relay.

2.2 Nano-machines Migration

When the concentration of molecules around the nano-machine is very high and reaches its own detection threshold H_i ($i = A, B_1, B_2, B_3$), the nano-machine moves under chemotaxis, and its motion follows the Langevin equation affected by external force $f(X, t)$. Nano-machines can move according to movement strategies in the system model.

$$\frac{d^2 X(t)}{dt^2} = -\alpha \frac{dX(t)}{dt} + \beta \frac{dW(t)}{dt} + f(X, t) \quad (3)$$

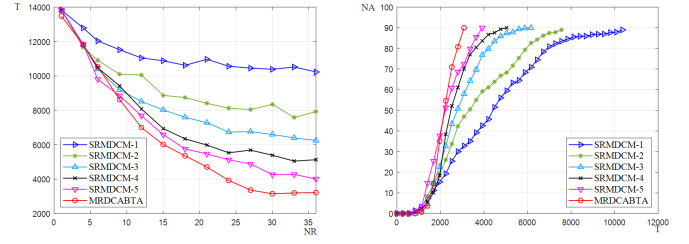
Algorithm 1 Nano-machine Migration Algorithm

```
while target not detected do
  if type A attractant detected then
    migrates towards the higher concentration of type A attractant
  else if type B1 attractant detected then
    migrates towards the higher concentration of type B1 attractant
  else if type B2 attractant detected then
    migrates towards the higher concentration of type B2 attractant
  else if type B3 attractant detected then
    migrates towards the higher concentration of type B3 attractant
  else if type B1 and B2 attractant detected then
    migrates towards the higher concentration of type B1 attractant
  else if type B2 and B3 attractant detected then
    migrates towards the higher concentration of type B2 attractant
  else if type B3 and B1 attractant detected then
    migrates towards the higher concentration of type B3 attractant
  else
    move in a random direction
  end if
end while
stop migrating
```

Algorithm 2 Relay Node Migration Algorithm

```
while target not detected do
  if The relay is moving then
    if type A attractant detected then
      stop migrating and release attractant B1
    else if type B1 attractant detected then
      stop migrating and release attractant B2
    else if type B2 attractant detected then
      stop migrating and release attractant B3
    else if type B1 and B2 attractant detected then
      stop migrating and release attractant B1
    else if type B2 and B3 attractant detected then
      stop migrating and release attractant B2
    else if type B3 and B1 attractant detected then
      stop migrating and release attractant B3
    else
      move in a random direction
    end if
  end if
end while
```

$$f(X, t) = \begin{cases} \gamma \nabla C(X, t) & C(X, t) \geq H_i \\ 0 & \text{otherwise} \end{cases} \quad (4)$$



(a) Efficiency of searching targets by nano-machine (b) Success rate of nano-machine searching target

Figure 1

Combined with the above formula, we can get formula (5).

$$\frac{d^2 X(t)}{dt^2} = \begin{cases} -\alpha \frac{dX(t)}{dt} + \beta \frac{dW(t)}{dt} + \gamma \nabla C(X, t) & C(X, t) \geq H_i \\ -\alpha \frac{dX(t)}{dt} + \beta \frac{dW(t)}{dt} & C(X, t) < H_i \end{cases} \quad (5)$$

α is a parameter that produces resistance to the movement of the nano-machine, β is a parameter of the degree to which the nano-machine is affected by noise, γ is represents the influence parameter of the chemotactic force on the movement of the nano-machine, $X(t)$ is the coordinate of the nano-machine at t , and $W(t)$ denotes Wiener process.

3 SIMULATION AND CONCLUSION

we use the proposed model to investigate the performance of a three attractant based on multi-hop molecular nano-network for target detection. We test the performance of the algorithm by comparing with different hops in the traditional multi-hop relay model.

Figure 1(a) shows the comparison between the proposed algorithm (MRDCABTA) and the traditional multi-hop relay algorithm (SRMDCM) with one hop, two hops, three-hop, four hops and five hops. and the horizontal axis NR is the number of relay nodes. Quantity, the vertical axis T is the cumulative time it took for 90% of the nano-machines to have detected the target. Obviously, the algorithm is more efficient in searching targets for nano-machines.

Figure 1(b) shows the comparison of the algorithm MRDCABTA with the traditional multi-hop relay algorithms (SRMDCM). The horizontal axis T is the time, and the number of cycles is reflected in the algorithm. The vertical axis NA is the success rate of the nanomachine searching for the target with the increase of T. Obviously, this algorithm has a better success rate in searching targets for nano-machines.

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